



CE-408L CRYPTOGRAPHY AND NETWORK SECURITY PROJECT PROPOSAL

SECTION: A **BATCH:** 2022F **NO. OF STUDENTS:** 4

GROUP MEMBERS

No.	Student Name	Roll Number	Signature
1.	Khalid Hussain	2022F-BCE-130	
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PROJECT TITLE

NarrativeIQ: A Data Warehousing and Big Data Platform for
Narrative Intelligence

PROJECT DESCRIPTION

NarrativeIQ is a Data Warehousing and Big Data project designed to analyze how public narratives form, grow, compete, and influence people over time. Instead of only showing simple trends, the system studies digital conversation signals such as events, topics, entities, sentiment, source activity, and relationships, then converts them into structured intelligence through an ETL pipeline and a warehouse-backed dashboard.



The main goal of NarrativeIQ is to help users understand not only what is trending, but why it is trending, which entities are involved, how public sentiment is changing, and what direction a narrative may take next. The system is built around a star-schema data warehouse, fact and dimension tables, materialized analytical views, API-ready marts, and an interactive dashboard.

In the intended full version, NarrativeIQ will collect data from multiple public information sources such as news, social platforms, public discussion forums, web references, and uploaded datasets. The data will be cleaned, transformed, enriched, stored in PostgreSQL, and analyzed through narrative intelligence modules.

Key features planned for the project include:

- Event and narrative intelligence dashboard.
- Topic IQ search for generating intelligence briefs on any topic.
- Narrative lifecycle tracking from emergence to growth, peak, decline, or mutation.
- Sentiment analysis across time and events.
- Entity intelligence for people, organizations, products, groups, and locations.
- Knowledge graph for entity and narrative relationships.
- Narrative comparison engine for side-by-side analysis.
- Narrative replay mode to show historical evolution.

- Prediction center for forecast-style narrative movement.
- Data quality dashboard for warehouse reliability.
- Branded reports in HTML, PDF, and CSV formats.
- Administration controls for ETL, warehouse loading, and audit logs.

NarrativeIQ demonstrates core Data Warehousing and Big Data concepts by using high-volume records, ETL processing, dimensional modeling, historical snapshots, data marts, data quality profiling, PostgreSQL loading, and analytical visualization. The project is suitable for an exhibition because it combines technical warehouse design with a clear real-world use case: understanding how narratives influence public perception.

		REMARKS
ACCEPTED	☐	
REJECTED	☐	

INSTRUCTOR DECISION

Ms. Arifa